

Informe de progreso técnico: Sistema de comercio electrónico OMMA (Proyecto MedBay)

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Introduction:

This report details the development activities carried out during the week of 1 to 5 December 2025. Following the structural upgrades implemented in previous weeks (categories, batch management, and country unification), this week's focus was on the reengineering and adaptation of the Bulk Import Module.

Although this module was already functional, it was necessary to carry out a deep update of its internal logic to align it with the new architecture of the system. The objective was to integrate the recent currency conversion capabilities (normalization to USD) and improve supplier management, in addition to implementing visual improvements and auditing tools (history) to ensure clean data entry consistent with the new reality of the system.

Development:

During the week of December 1 to 5, integration **and refactoring took place**. Although the "Mass Loading" module was already functional in the early stages of the project, the significant evolution of the core modules (Countries, Inventory, Categories) during the last month generated a "technical debt" in the import process. The old logic was no longer supported by the new data structure.

Therefore, this week's work was not simply a repair, but a

Complete reengineering of the data entry flow, ensuring that the information entered respects the new global business rules, especially in financial and geographical terms.

A. Reengineering and Logical Adaptation

Updating the database schemas in the Inventory and Products modules required the import engine to change the way it processed Excel files.

Architecture Synchronization: Rewrote the file reading algorithm to align with the new relational tables.

The Problem: Having modified the way products are created (now with strict categories) and batches (now with global visualization), the old importer failed or left data unconnected.

The Solution: Updated the backend logic to ensure referential integrity. Now, when uploading a file, the system validates and distributes the data correctly towards the new table structures, ensuring that an imported product is immediately visible and functional in the updated Inventory and Categories modules.



B. Financial and Geographic Integration (Currency Translation)

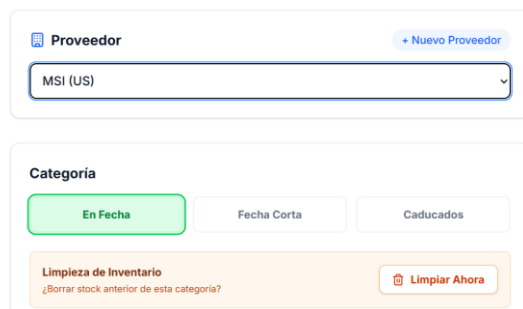
This is the most strategic functional advancement of the week. The Countries Module (developed three weeks ago) was connected directly to the Bulk Import flow.

Real-Time Monetary Normalization: Implemented logic to handle currency diversity from entry.

Process Flow: When uploading a catalog, the system now queries the "Value Rate" (Exchange Rate) associated with the supplier's country.

Automatic Conversion: If a supplier is from Mexico, the system identifies the MXN/USD rate. When reading the prices from Excel (which come in Pesos), the system performs an automatic mathematical conversion before saving the record in the database.

Result: All inventory, regardless of whether it comes from suppliers in Europe, Asia or Latin America, is stored in U.S. Dollars (USD). This allows for unified financial reporting and eliminates the complexity of handling multiple currencies in sales views.



C. Supplier Management Optimization (UI/UX)

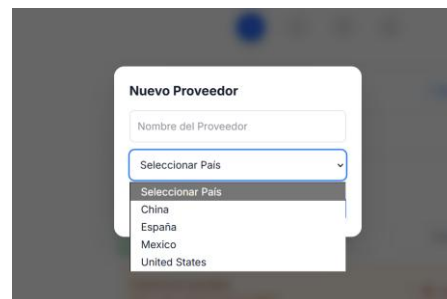
The user interface has been redesigned to provide clarity on assigning ownership to catalogs.

Separation of Flows (Selection vs. Creation): Visually, distinct and clear sections were created to avoid operational errors:

Existing Supplier: A simplified flow to update stock of already registered partners.

New Provider: A dedicated form that demands critical data. Now, when creating a supplier from the import, it is **mandatory to select its Name and Country of Origin**. This "Country" selection is the trigger that enables the currency conversion mentioned in point B.

Visual Improvements: Modern design principles ("Visual Touches") were applied to improve the readability of the process. The use of guided steps (wizards) and visual validations reduces the cognitive load of the administrator, making the loading process less prone to human error.



With these implementations, the Bulk Upload module has ceased to be an isolated tool and has become a fully integrated component of the MedBay intelligent ecosystem.

Conclusions:

Updating the import module closes the development loop of the administrative backend. The data "gateway" now speaks the same language as the rest of the system, ensuring that referential integrity is maintained from Excel upload to final inventory.

The successful integration of the Countries module into the import process is a key milestone: the system is now truly capable of operating globally, accepting catalogs in various local currencies but financially standardizing inventory in Dollars. In addition, the separation of flows for the creation of suppliers and the implementation of load history provide the administrator with control and traceability that did not exist before, significantly reducing the risk of operational errors.

References:

1. **Oracle Help Center (2025).** *Multi-Currency Accounting and Inventory Management*. Retrieved from <https://docs.oracle.com/> (Standards for the conversion and management of inventories in base/foreign currency).

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